SOFTWARE ENGINEERING LAB

LAB 6

Design Analysis Report

Digital Document Verification System

Name: Ansh Tyagi

Roll no :202401100200058

Aim	To study the design of a digital document verification system, including its main components, requirements, database, and constraints.
Project	A web-based system for uploading documents and tracking verification results.

The following report describes a proposed design. Specific technologies can be selected during implementation.

1. Introduction

The Digital Document Verification System is intended to make document submission and verification easier. A user can upload a document, such as a certificate or identity proof, and check its status later. An administrator or verifier reviews the submitted document and records the result. This approach can reduce paperwork and make it easier to keep track of pending and completed requests.

2. Objectives

- Allow users to submit documents online.
- Check that uploaded files meet basic format and size requirements.
- Give authorized verifiers a place to review submitted documents.
- Record the result as Pending, Verified, or Rejected.
- Let users check the status of their own submissions.
- Inform users when a verification decision has been made.

3. System Design and Architecture

The system can be divided into three parts. Keeping these responsibilities separate makes the application easier to understand and maintain.

Part of system	Main responsibility
Presentation layer	The web pages used by users and administrators, including upload forms and status screens.
Application layer	Login checks, file validation, verification workflow, status updates, and notifications.
Data layer	Stores user details, document information, verification decisions, and related records.

In a typical flow, the user uploads a file through the interface. The application checks the submission and saves its details. A verifier then reviews the document and the application records the final status.

4. Functional Requirements

No.	Requirement
1	Users should be able to register and log in.
2	Users should be able to upload a document with the required details.
3	The system should check file format, size, and required fields.
4	Authorized verifiers should be able to review pending submissions.
5	The system should save the verification decision and update the status.
6	Users should be able to view the status of their own documents.
7	The system should show or send a notification when the status changes.

5. Non-Functional Requirements

Security: Only authorized people should be able to view or manage protected documents.

Performance: Uploading a file and checking a status should take a reasonable amount of time.

Reliability: The system should save verification decisions correctly and handle errors without losing records.

Usability: The pages and status messages should be simple enough for a first-time user to understand.

Maintainability: The code should be divided into clear modules so changes are easier to make.

Scalability: The design should allow the system to handle more users and documents as usage grows.

6. Database Design

The following tables or collections could be used. The exact structure may change depending on the database selected.

Entity	Example fields	Use
User	userId, name, email, passwordHash, role	Stores account and role details.
Document	documentId, userId, fileUrl,	Stores document information and

	documentType, status, uploadedAt	status.
Verification	verificationId, documentId, verifierId, remarks, verifiedAt	Records the review and decision.
Notification	notificationId, userId, message, createdAt, isRead	Tracks status notifications.

7. Design Constraints

- Only logged-in and authorized users should access protected features.
- The system should limit file types and sizes to reduce invalid or unsafe uploads.
- A user should not be able to view another user's private documents.
- Verification decisions should include the reviewer and the time of the decision.
- The application should display a useful message if an upload or database request fails.

8. Feasibility Analysis

Technical feasibility: The project can be built with common web-development tools, a backend API, and a database.

Economic feasibility: Using open-source frameworks can keep development costs low.

Operational feasibility: A browser-based interface should make document submission and status checking straightforward.

Security considerations: Authentication, role permissions, secure storage, and activity logs should be included in the implementation.

9. Suggestions for Improvement

- Store actual files in protected file storage and keep file references and metadata in the database.
- Use role-based permissions for users and verifiers.
- Keep a record of status changes so decisions can be reviewed later.
- Validate uploaded files on the server, not only in the browser.
- Provide clear error messages and allow safe retries when a request fails.

10. Expected Outcome

The proposed system should give users a convenient way to submit documents and check their status. It should also help administrators manage pending requests and keep verification decisions in one place.

11. Conclusion

In this lab, the design of a Digital Document Verification System was analyzed. The main requirements, system layers, database entities, constraints, and feasibility were identified. This analysis can be used as a starting point for developing and testing the actual application.